Which mistakes get punished at this level?

A human-move model's odds of the punishing reply, checked on the player's own games

Chess Coach research lab

Abstract

An engine prices a mistake by what it would cost against perfect play. At the player's level, many mistakes cost nothing, because the reply that punishes them is never found. Maia-2, the human-move model the coaching app ships, gives the probability that a player at a given rating plays each move. Before building on it, we asked whether its probability of the punishing reply can be trusted on the player's own games. Over 689 errors in 60 games, by a rule fixed before any number was computed, it can: its odds beat both a constant base rate and a fit on the size of the error, and every bin of its odds is calibrated, for the player's errors (which the opponents punish) and for the opponents' (which the player punishes), on verified replies and on all of them. Of the player's errors that cost at least 10 points of winning chances, 45% went unpunished, and the odds tell the punished from the unpunished with an area under the ROC curve of 0.80. The app now prices every error with these odds. The lab recomputes the analysis byte for byte from the recorded events, and a rerun on engine evidence rebuilt on the app as it now runs reaches the same verdicts.

1 The question

A review that judges a mistake only by the engine treats every hanging piece alike. At the player's level that is not how games go: some mistakes are punished at once, and some survive because the punishing reply is hard to see. A verdict that weighs a mistake *given the circumstance* needs to know how likely the punishment was. The cheapest circumstance to measure is the reply itself: did the opponent find the move that punishes the error?

Maia-2 is a neural network trained on human games that gives, for a position and a rating band, the probability of each legal move. The app runs it in Node through its engine service. If its probability of the punishing reply is trustworthy on this player's games, a review can say how often a player at the opponent's level finds the punishment, and the daily coach can rank the player's mistakes by how likely they are to be punished.

2 Events

Every error in the 60 games of the accuracy project: a move that gave away at least 5 points of the mover's winning chances against the engine's best move, read from the routine-profile reviews. Each error is an event with:

- the **punishing reply**: the engine's best move in the position after the error;
- **E1**: the reply played was exactly that move (the outcome the rule scores);

- **E2**: the reply played gave away at most 2 points (an equally good alternative counts; a secondary outcome);
- ***q***: Maia-2's probability of the punishing reply at the replier's band (1000 to 1400 in these games), read from the replier's chess.com rapid rating.

The events run in two directions: the player's errors, which the opponents reply to, and the opponents' errors, which the player replies to. A reply from the opening book is left out (its loss is a placeholder). The primary set keeps the 517 replies whose evidence the review's deeper pass verified, which is how the question reads "the engine's best reply". Because that pass checks errors first, the set is selected partly on the outcome, so every result is also given on all 689 non-book replies.

3 The rule

Fixed before any number was computed: Maia is trusted in a direction when (1) its Brier score for E1 beats both a constant base rate and a logistic regression on the error's size alone, both fitted leaving one game out, with the 95% interval of each difference below zero over 1000 bootstrap draws of games; and (2) no calibration bin of q with 15 or more events has its mean q more than 0.1 outside the Wilson interval of the share that was found.

4 Results

The rule holds in both directions and on both sets (Table 1). On the player's errors, Maia's Brier score is 0.162 against 0.249 for the base rate and 0.236 for the fit on the error's size; on the opponents' errors, 0.145 against 0.247 and 0.231. The secondary outcome is weaker: on all replies to the player's errors, Maia does not beat the base rate for E2 (difference −0.001, [−0.031, 0.031]), since an equally good alternative is often a different move.

replies	errors of	events	punished	q	base	size	q − base	q − size
verified	player	243 (57)	0.436	0.162	0.249	0.236	−0.087 [−0.118, −0.056]	−0.074 [−0.106, −0.046]
verified	opponents	274 (59)	0.427	0.145	0.247	0.231	−0.102 [−0.125, −0.079]	−0.087 [−0.113, −0.060]
all	player	334 (57)	0.419	0.162	0.245	0.241	−0.083 [−0.105, −0.060]	−0.079 [−0.103, −0.055]
all	opponents	355 (60)	0.425	0.151	0.246	0.238	−0.095 [−0.116, −0.074]	−0.087 [−0.110, −0.066]

Table 1: Brier scores for E1 (the punishing reply is played): Maia's odds (q), the base rate (*base*) and the fit on the error's size (*size*), both leaving one game out, and the paired differences with 95% bootstrap intervals over games. Events with games in parentheses; *punished* is the share of E1.

Calibration. Every bin is inside the rule (Table 2, Figure 1). One bin is flagged by a stricter reading (its mean q outside the Wilson interval and more than 0.1 from what was found): the player punishes the opponents' errors less often than Maia expects where q is 0.2 to 0.4 (0.15 found against a mean q of 0.29, $n = 53$). It is inside the rule, and worth watching as games accumulate.

	the player's errors			the opponents' errors		
q	mean q	found	n	mean q	found	n
0.0–0.1	0.04	0.09	58	0.04	0.09	55
0.1–0.2	0.15	0.20	25	0.15	0.18	38
0.2–0.4	0.31	0.30	47	0.29	0.15	53*
0.4–0.6	0.50	0.57	30	0.51	0.58	43
0.6–0.8	0.71	0.70	40	0.72	0.76	37
0.8–1.0	0.91	0.86	43	0.90	0.92	48

Table 2: Calibration on verified replies: by bin of q, its mean, the share of punishing replies found, and the number of events. * flagged by the stricter reading.

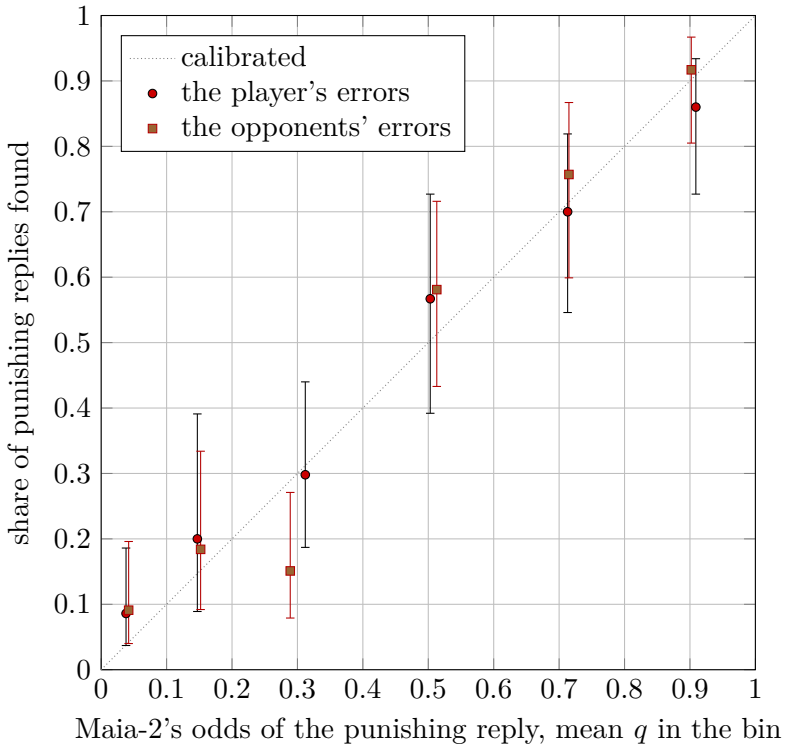
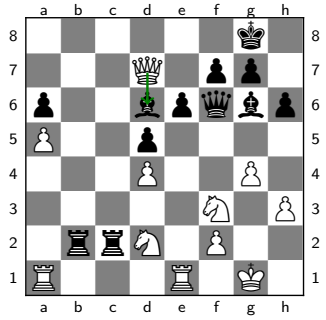


Figure 1: Calibration on verified replies, with 95% Wilson intervals.

5 What it means for verdicts

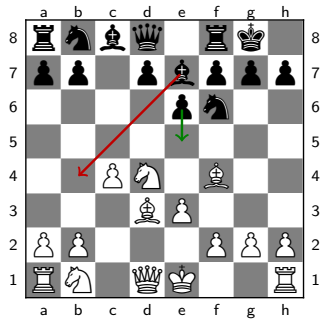
Of the errors that cost at least 10 points, 45% of the player's (140 errors) and 41% of the opponents' (157) went unpunished in these games: the reply played gave away more than 2 points. The odds tell them apart. The mean q was 0.62 for the player's errors that were punished and 0.28 for those that were not, and the area under the ROC curve is 0.80 [0.72, 0.88] for the player's errors and 0.81 [0.73, 0.88] for the opponents' (on all replies, 0.76 and 0.80).

Figures 2 and 3 show two of the player's errors that cost at least 20 points, on verified replies (58 such errors): the punished one Maia was surest of, other than taking back the piece that just moved, and the unpunished one Maia was surest would be missed. The green arrow is the punishing reply; the red one, the reply played when it was something else.



The error: 21…♖cc2, giving away 22.6% of the player's winning chances.
The punishing reply: 22. ♕xd6 (green), the engine's best; Maia-2 at 1200 gives it $q = 0.954$.
Played: 22. ♕xd6, the punishing reply.
UCI: error `c8c2`; punishing reply `d7d6`; played `d7d6`
FEN: `6k1/3Q1pp1/p2bpqbp/P2p4/3P2P1/5N1P/1rrN1P2/R3R1K1 w - - 2 22`

Figure 2: Found: Maia-2 gives 22. ♕xd6 $q = 0.954$ at 1200, and the opponent played it.



The error: 7. ♘xd4, giving away 24.1% of the player's winning chances.
The punishing reply: 7…e5 (green), the engine's best; Maia-2 at 1300 gives it $q = 0.003$.
Played: 7…♗b4+ (red), which gave away 27.1% of the replier's winning chances.
UCI: error `f3d4`; punishing reply `e6e5`; played `e7b4`
FEN: `rnbq1rk1/pp1pbppp/4pn2/8/2PN1B2/3BP3/PP3PPP/RN1QK2R b KQ - 0 7`

Figure 3: Missed: Maia-2 gives 7…e5 $q = 0.003$ at 1300, and the opponent played 7…♗b4+ instead.

In the app. The review service now prices every error of at least 5 points with these odds, computed as here (the replier's band, the engine's best answer). Game Review shows the line under a mistake, and the routine's teaching pass picks its deep looks by the winning chances lost times the odds. A turning point with no odds of its own is ranked at 0.44, the share of the player's errors punished in this check.

6 Reproduction in the lab

Recomputed. The lab runs the analysis (`analyze.mjs`) over the events the records keep, and its results matches the records' results byte for byte. The records are read at `b3b827d`.

Rerun. The lab rebuilt the routine evidence for the same games on the app as it now runs (`d8b7712`), found the errors and their punishing replies again, ran Maia-2 on each, and reran the analysis: 684 events against 689 recorded, since the rebuilt evidence differs in detail. Every verdict holds again; no Brier score moves by more than 0.004 and no area under the curve by more than 0.04 (Table 3).

		events		Brier, q		AUC		verdict	
replies	to errors of	recorded	rerun	recorded	rerun	recorded	rerun	recorded	rerun
verified	the player's	243	245	0.162	0.160	0.80	0.76	holds	holds
verified	the opponents'	274	276	0.145	0.146	0.81	0.85	holds	holds
all	the player's	334	336	0.162	0.163	0.76	0.72	holds	holds
all	the opponents'	355	348	0.151	0.147	0.80	0.83	holds	holds

Table 3: The recorded analysis beside the rerun on rebuilt evidence. AUC is for E2 on errors of at least 10 points.

7 Limitations

- 60 games of one player and their opponents, all near the same rating. The check says Maia can be trusted at this level, not at every level.
- E1 counts only the engine's best move as the punishment. A reply that punishes as well by another route counts as missed under E1 and found under E2.
- The band comes from a conversion of the chess.com rating, and a player's form on the day is not in it.